

Nicholas Mayer-Rupert

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EDUCATION

Simon Fraser University

Bachelor of Science in Computer Science

Vancouver, BC

Sep 2022 – Jun 2026

SKILLS

Languages: Java, C++, Python, JavaScript, TypeScript, SQL

Back end: Node.js, Spring Boot, Fastify, PostgreSQL, Firebase, WebSocket

Front end: React, Tailwind CSS, Three.js, WebGL2, Mapbox GL

Systems & other: WebAssembly, Git, Cloudflare, OAuth2, Twilio, CSV/ETL

EXPERIENCE

Mastercard

Software Engineer I

Aug 2026 – Present

Connection Lab

Full-Stack Developer Intern

Hybrid – Vancouver

May 2025 – Dec 2025

- Shipped a Node.js and PostgreSQL platform for cohort wellness with invite-based signup, de-duplicated check-ins, threaded comments, progress visualizations, moderation, and admin tools.
- Built an inline-CRUD facilitator dashboard with cross-table search across organizations, cohorts, participants, check-ins, and comments.
- Migrated operational data from Airtable to PostgreSQL and implemented email reset, Google Sign-In, and role-based access.
- Implemented replies and SMS notifications with quiet hours and per-user limits.

PROJECTS

Falling Sand Engine | C++, WebAssembly, WebGL2, React

2026 – Present

- Built a two-layer falling-sand simulation with procedural infinite terrain, rigid bodies, lighting, creatures, survival progression, and multiplayer.
- Moved simulation, rendering, camera, input policy, tools, and streaming into a composed C++ engine compiled to WebAssembly; React mounts a framework-free web component.
- Designed a worker-authoritative world with compact diffs, a main-thread presentation mirror, and deterministic local-player prediction.
- Created benchmark and checksum suites to isolate optimizations while preserving exact behavior; hosted at nicholasmayerrupert.com/work/falling-sand.

StoryCarbon | React, TypeScript, Mapbox, Firebase, Tailwind CSS

Jan – Apr 2024

- Built an interactive Mapbox experience for sponsoring farmland plots used for tree planting, with a responsive React and Tailwind CSS interface.
- Integrated Firebase authentication, storage, and real-time data, and surfaced impact in metric tons of CO₂ sequestered.
- Presented StoryCarbon at a BioGenerator meeting in St. Louis for venture-capital funding.

3D Conway's Game of Life | React, Three.js, C++, WebAssembly

2024 – Present

- Visualized a finite 16×16 toroidal Game of Life board by stacking each generation as a Three.js layer along the vertical axis.
- Built a parallel C++/WebAssembly soup-search engine and searched hundreds of millions of seeds for rare long-lived patterns.
- Found and published a non-trivial 5,024-frame attractor reached after 176 transient generations, with interactive editing and playback controls.